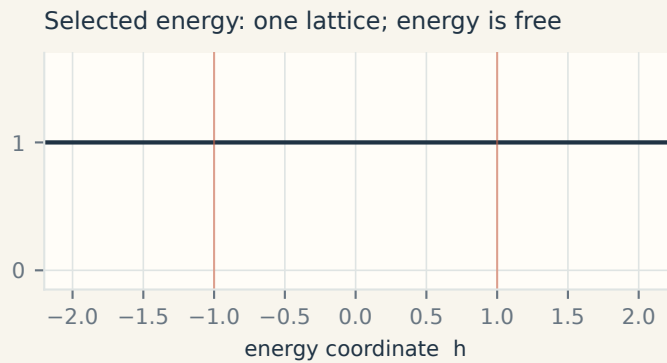
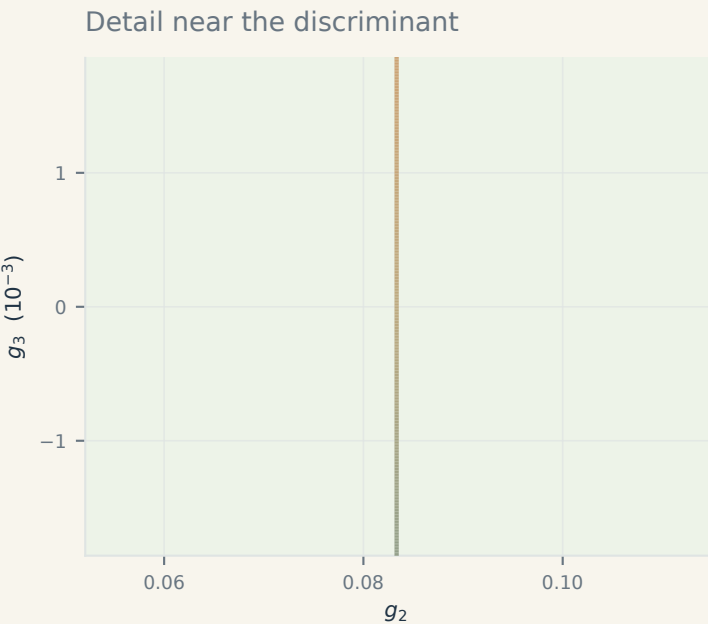
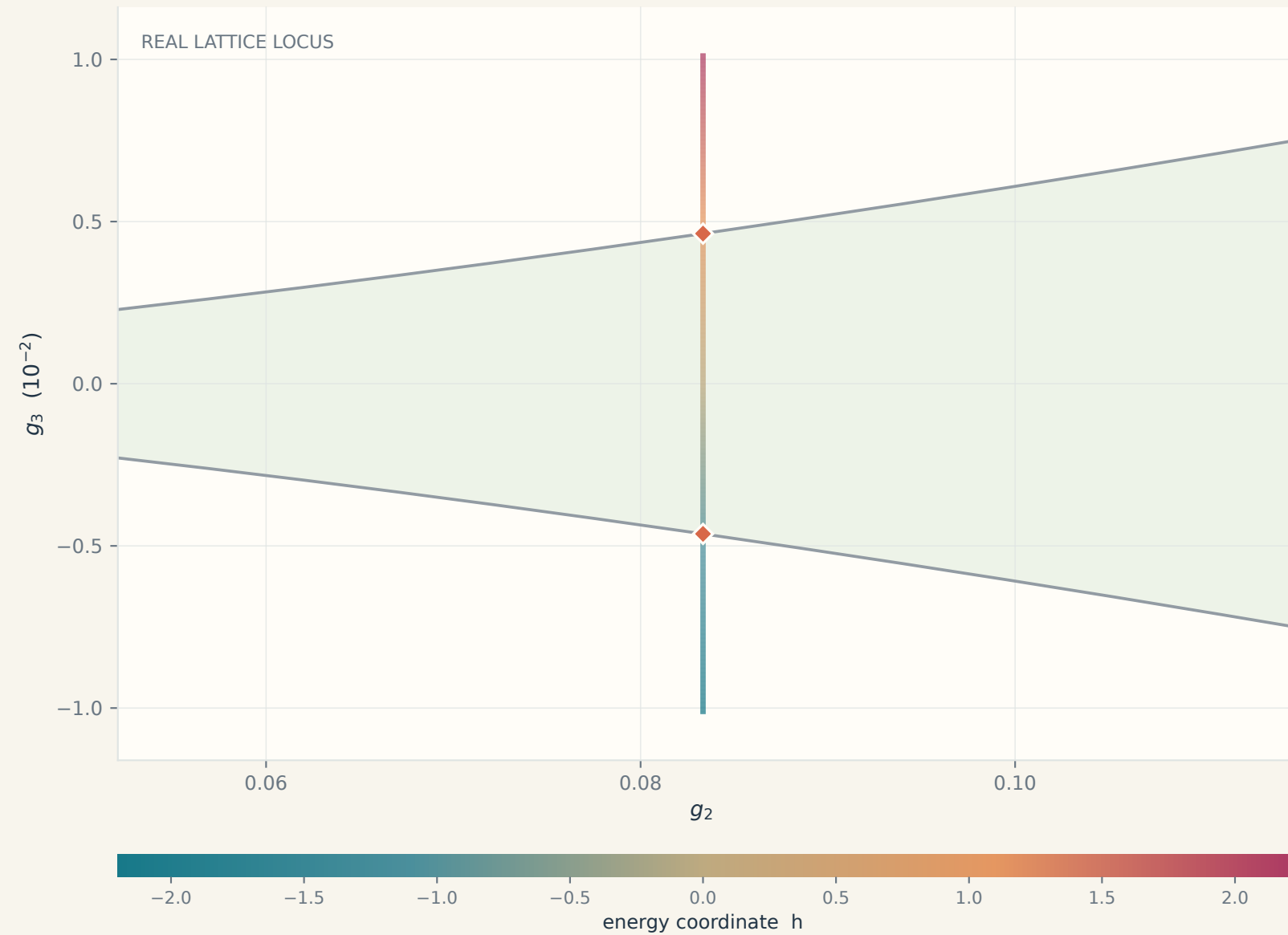


KdV · the free-energy line

$u'' - u - 6u^2 = 0$

$(n, p) = (2, 2) \cdot q = 2$



Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

One fixed equation carries a continuum of lattices; $h = 216 g_3$ is its first-integral parameter.
The endpoints $h = \pm 1$ are degenerations. The open segment between them has $\Delta > 0$.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.